DEPARTMENT OF SMART COMPUTING AND CYBER RESILIENCE

FACULTY OF ENGINEERING AND TECHNOLOGY

FINAL PROJECT FOR CSC1024\PRG3024\FEL1184 PROGRAMMING PRINCIPLES

ACADEMIC SESSION: September 2025

DISTRIBUTION: WEEK 06

SUBMISSION DEADLINE: 19th December 2025 (Friday), before 11.59 PM.

Student Names with IDs:

1. _____
   (Group Leader)
2. _____
3. _____
4. _____
5. _____

INSTRUCTIONS TO CANDIDATES

- This programming project will contribute **50%** to your final grade.
- This programming project is a final examination.
- This is a group project with a maximum of **5** members.
- The evaluation will be carried out in **Week 13** during lab slots (Please email your lab instructor to book your evaluation slot)
- The group leader will submit the following materials via elearn :
  - ✓ Report in .pdf
  - ✓ Contribution log signed by each member of the group
  - ✓ Program source file in either .py or .ipynb
  - ✓ Link to recorded presentation
  - ✓ Presentation deck in .pdf
- Please ensure that for both report and presentation deck submitted, each student name and ID are included. Also, do note that marks for the project may differ from each member based on their understanding and contribution to the project.

IMPORTANT

The University requires students to adhere to submission deadlines for any form of assessment. Penalties are applied in relation to unauthorized late submission of work.

- Project submitted after the deadline but within **1 week** will be accepted for a maximum mark of **40%**.
- Work handed in following the extension of **1 week** after the original deadline will be regarded as a non-submission and marked **ZERO**.

Lecturer's Remark (Use additional sheet if required)

We, the students as presented **above**, have received the programming project and read the comments.

……..……..…..

(Student signatures and Date)

Academic Honesty Acknowledgement

We, the students as presented above, verify that this paper contains entirely our own work. We have not consulted with any outside person or materials other than what was specified (such as an interviewee) in the assignment or the syllabus requirements. Further, we have not copied or inadvertently copied ideas, sentences, or paragraphs from another student. We understand the penalties (as outlined in the student handbook) for any form of copying or collaboration on any assignment.

……..…….....

(Student signatures and Date)

CSC1024 Programming Principles
Final Project

Python-Powered Student Grading System

Project Description:

The goal of this project is to develop a Student Grading System that manages student records and their assessment results. The program should calculate overall grades, assign letter grades, and display class performance summaries.

Project Requirements:

a. **Initial Data Preparation:** Create the following text files:
   - ✓ students.txt – Student ID, name, and email.
   - ✓ grades.txt – Student ID, course name, marks, and grade.
   - ✓ courses.txt – Course ID and course name.

b. **Program Functionality:** Implement a user-friendly menu system with the following options (atleast):
   - ✓ Add a new student: Allow users to register a new student by providing details such as Student ID, name, and email address.
   - ✓ Add a new course: Enable users to add a new course to the system by providing the Course ID and course name.
   - ✓ Record student marks: Allow users to enter marks for a student in a specific course, automatically calculating and assigning a letter grade.
   - ✓ Display individual student performance: Provide an option to display the performance report of a student, including all enrolled courses and grades.
   - ✓ Display course performance summary: Display the list of students enrolled in a particular course, along with average, highest, and lowest marks.
   - ✓ Export performance report: Enable exporting of the summary or individual performance report to a text file for record keeping.
   - ✓ Exit: Provide an option to exit the program gracefully.

c. Programming Techniques:

- ✓ Use Command Line Interfacing.
- ✓ Use input and display functions for user interaction and data input/output.
- ✓ Employ file handling for data persistence and retrieval.
- ✓ Use conditional statements and loops for decision-making and repeated tasks.
- ✓ Define functions for modularity and reusability.

Marking Rubrics (Total Marks: 100):

Group Evaluation (70%)

a. Code Organization and Readability (5 marks):

- ✓ Proper indentation and consistent coding style (2 marks)
- ✓ Meaningful variable and function names (1 marks)
- ✓ Appropriate use of comments to enhance code understanding (1 marks)
- ✓ Logical and efficient code structure (1 marks)

b. Functionality and Correctness (25 marks):

- ✓ Accurate implementation of all main functionalities (5 marks)
- ✓ Correct data validation and logical operations (5 marks)
- ✓ Proper handling of updates and deletions (10 marks)
- ✓ Display of information with accuracy and clarity (5 marks)

c. Data Persistence and File Handling (10 marks):

- ✓ Proper creation and utilization of text files for data storage (5 marks)
- ✓ Accurate reading and writing of data to and from files (5 marks)

d. User Interface and User Experience (8 marks):

- ✓ User-friendly menu system (2 marks)
- ✓ Clear and concise prompts and messages (2 marks)
- ✓ Smooth navigation and flow of the program (4 marks)

e. Error Handling and Robustness (8 marks):

- ✓ Appropriate error handling for input validation (4 marks)
- ✓ Graceful handling of edge cases and exceptions (2 marks)
- ✓ Overall program stability and reliability (2 marks)

f. Flowchart (4 marks):

- ✓ Clarity and readability of the flowchart (1 mark)
- ✓ Accurate representation of the program flow (1marks)
- ✓ Inclusion of all major processes and decision points (1 mark)
- ✓ Proper use of flowchart symbols and conventions (1 mark)

g. Report (10 marks):

- ✓ Clarity and organization of the report (2 marks)
- ✓ Detailed explanation of the system design and architecture (2 marks)
- ✓ Discussion of the implementation challenges and solutions (2 marks)
- ✓ Analysis of the system's strengths and limitations (2 marks)
- ✓ Suggestions for future improvements and enhancements (2 marks)

Individual Evaluation (30%)

h. Presentation (10 marks):

- ✓ Clarity and effectiveness of the presentation (2 marks)
- ✓ Demonstration of the system's functionality (3 marks)
- ✓ Explanation of the key features and design decisions (3 marks)
- ✓ Time management and adherence to the allotted presentation duration (2 mark)

i. Contribution (20 marks):

- ✓ Code Contribution: Student’s actual code ownership, logic accuracy, and integration (4 marks)
- ✓ Understanding of Code: Ability to explain and justify written code during viva/reflection (8 marks)
- ✓ Team Collaboration: Participation, communication, and contribution logs (3 marks)
- ✓ A Professionalism & Responsibility: Commitment, documentation effort, and ethics (2 marks)